

# SmoothL1Loss#

<https://pytorch.org/docs/stable/generated/torch.nn.SmoothL1Loss.html>

## Description:

Creates a criterion that uses a squared term if the absolute element-wise error falls below  $\beta$  and an L1 term otherwise. It is less sensitive to outliers than `torch.nn.MSELoss` and in some cases prevents exploding gradients (e.g. see the paper Fast R-CNN by Ross Girshick). For a batch of size  $N$ , the unreduced loss can be described as: If reduction is not `None`, then:

## Function Signature

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```
class torch.nn.SmoothL1Loss(size_average=None, reduce=None,
                             reduction='mean', beta=1.0)[source]#
```

Parameters

Shape:

```
forward(input, target)[source]#
```

Return type

## Returns:

Tensor

## Notes & Warnings

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### NOTE

Note Smooth L1 loss can be seen as exactly L1Loss, but with the  $|x - y| < \beta$  portion replaced with a quadratic function such that its slope is 1 at  $|x - y| = \beta$ . The quadratic segment smooths the L1 loss near  $|x - y| = 0$ .

### NOTE

Note Smooth L1 loss is closely related to HuberLoss, being equivalent to  $\text{huber}(x, y) / \beta$  (note that Smooth L1's  $\beta$  hyper-parameter is also known as  $\delta$  for Huber). This leads to the following differences: As  $\beta \rightarrow 0$ , Smooth L1 loss converges to L1Loss, while HuberLoss converges to a constant 0 loss. When  $\beta$  is 0, Smooth L1 loss is equivalent to L1 loss. As  $\beta \rightarrow +\infty$ , Smooth L1 loss converges to a constant 0 loss, while HuberLoss converges to MSELoss. For Smooth L1 loss, as  $\beta$  varies, the L1 segment of the loss has a constant slope of 1. For HuberLoss, the slope of the L1 segment is  $\beta$ .

## Detailed Documentation

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## Documentation

Creates a criterion that uses a squared term if the absolute element-wise error falls below  $\beta$  and an L1 term otherwise. It is less sensitive to outliers than `torch.nn.MSELoss` and in some cases prevents exploding gradients (e.g. see the paper Fast R-CNN by Ross Girshick).

For a batch of size  $N$ , the unreduced loss can be described as:

If reduction is not none, then:

Smooth L1 loss can be seen as exactly L1Loss, but with the  $|x - y| < \beta$  portion replaced with a quadratic function such that its slope is 1 at  $|x - y| = \beta$ . The quadratic segment smooths the L1 loss near  $|x - y| = 0$ .

Smooth L1 loss is closely related to HuberLoss, being equivalent to  $\text{huber}(x, y) / \beta$  (note that Smooth L1's  $\beta$  hyperparameter is also known as  $\delta$  for Huber). This leads to the following differences:

As  $\beta \rightarrow 0$ , Smooth L1 loss converges to L1Loss, while HuberLoss converges to a constant 0 loss. When  $\beta$  is 0, Smooth L1 loss is equivalent to L1 loss.

As  $\beta \rightarrow +\infty$ , Smooth L1 loss converges to a constant 0 loss, while HuberLoss converges to MSELoss.

For Smooth L1 loss, as  $\beta$  varies, the L1 segment of the loss has a constant slope of 1. For HuberLoss, the slope of the L1 segment is  $\beta$ .

`size_average` (bool, optional) – Deprecated (see `reduction`). By default, the losses are averaged over each loss element in the batch. Note that for some losses, there are multiple elements per sample. If the field `size_average` is set to False, the losses are

instead summed for each minibatch. Ignored when reduce is False. Default: True

reduce (bool, optional) – Deprecated (see reduction). By default, the losses are averaged or summed over observations for each minibatch depending on size\_average. When reduce is False, returns a loss per batch element instead and ignores size\_average. Default: True

reduction (str, optional) – Specifies the reduction to apply to the output: 'none' | 'mean' | 'sum'. 'none': no reduction will be applied, 'mean': the sum of the output will be divided by the number of elements in the output, 'sum': the output will be summed. Note: size\_average and reduce are in the process of being deprecated, and in the meantime, specifying either of those two args will override reduction. Default: 'mean'

beta (float, optional) – Specifies the threshold at which to change between L1 and L2 loss. The value must be non-negative. Default: 1.0

Input:  $(*)^{(*)}(*)$ , where  $*^{**}$  means any number of dimensions.

Target:  $(*)^{(*)}(*)$ , same shape as the input.

Output: scalar. If reduction is 'none', then  $(*)^{(*)}(*)$ , same shape as the input.

Runs the forward pass.